

Amir Hossain Raj

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RESEARCH STATEMENT

I am broadly interested in building intelligent robotic systems that can perceive, learn, and act in complex real-world environments. My research focuses on reinforcement learning, generative AI, and computer vision, with applications in navigation and motion planning for autonomous mobile robots. I am also interested in dexterous locomotion and manipulation, as well as long-horizon task and motion planning that enables robots to solve multi-step problems safely and efficiently.

EDUCATION

Doctor of Philosophy in Computer Science 2022 - Present
George Mason University, VA, USA

Master of Science in Computer Science 2022 - 2024
George Mason University, VA, USA
CGPA: 3.82/4.00

Bachelor of Science in Computer Science and Engineering 2016 - 2021
Ahsanullah University of Science and Technology, Dhaka
Thesis Title: Natural Language Video Captioning in Bengali Using Deep Learning
CGPA: 3.85/4.00

WORK EXPERIENCE

Graduate Research/Teaching Assistant, George Mason University Aug 2022 - Present

- Developed and implemented reinforcement learning and vision-language models for robot navigation and locomotion in complex environments. Focus on multimodal perception, motion planning algorithms for mobile robots, and deployment of autonomous systems.
- Supported teaching in programming and data analysis courses by helping students understand high-level and object-oriented programming concepts. Assisted with assignments, debugging, and practical problem-solving. Provided help through office hours, grading, and direct student feedback.

Lecturer, Ahsanullah University of Science and Technology Nov 2021 - Jul 2022

- Conducted lab sessions: Software Development, Compiler Design, Operating Systems. Guided diverse learners with patience, adaptability, and clear communication, fostering confidence and collaborative engagement.

Associate Software Engineer, Brain Station 23 Jan 2021 - Nov 2021

- Worked as a Machine Learning and AI Engineer on research problems including Image Forgery Detection, Automatic Radiology Report Generation, Medical Machine Reading Comprehension, and Unsupervised Clinical Trial Data Analysis. Workload included programming, design and development to launch products.

TECHNICAL SKILLS

Work experience with C++, Python, PyTorch, ROS, Isaac Lab etc.

Journal

1. Daeun Song, Jing Liang, Amirreza Payandeh, **Amir Hossain Raj**, Xuesu Xiao, Dinesh Manocha, “**VLM-Social-Nav: Socially Aware Robot Navigation through Scoring using Vision-Language Models**”, IEEE Robotics and Automation Letters (Volume: 10, Issue: 1, January 2025).
2. Saiful Islam, Aurpan Dash, Ashek Seum, **Amir Hossain Raj**, Tonmoy Hossain, Faisal Muhammad Shah, “**Exploring Video Captioning techniques: A Comprehensive Survey on Deep Learning Methods**”, SN Computer Science 2, no. 2 (2021): 1-28.
3. Al Jaber Mahmud, **Amir Hossain Raj**, Duc M. Nguyen, Xuesu Xiao and Xuan Wang, “**DARC: Disturbance-Aware Redundant Control for Human–Robot Co-Transportation**”, Electronics, vol. 14, no. 12, 2480, 2025. (MDPI Electronics).

Conference

1. Amirreza Payandeh, Daeun Song, Mohammad Nazeri, Jing Liang, Praneel Mukherjee, **Amir Hossain Raj**, Yangzhe Kong, Dinesh Manocha, Xuesu Xiao. Social-LLaVA: Enhancing Robot Navigation through Human-Language Reasoning in Social Spaces. (IROS 2025).
2. Al Jaber Mahmud, **Amir Hossain Raj**, Duc M Nguyen, Xuesu Xiao, Xuan Wang. Human-Robot Co-Transportation using Disturbance-Aware MPC with Pose Optimization. (IROS 2025).
3. **A. H. Raj**, Z. Hu, H. Karnan, R. Chandra, A. Payandeh, L. Mao, P. Stone, J. Biswas, and X. Xiao. Rethinking social robot navigation: Lever- aging the best of two worlds. (ICRA 2024).
4. Zifan Xu, **Amir Hossain Raj**, Xuesu Xiao, and Peter Stone. Dexterous Legged Locomotion in Confined 3D Spaces with Reinforcement Learning. (ICRA 2024).
5. **Amir Hossain Raj**, Ashek Seum, Aurpan Dash, Saiful Islam, Faisal Muhammad Shah, “**Deep Learning Based Video Captioning in Bengali**”, The 26th IEEE International Conference on Automation and Computing (ICAC’21).
6. Ashek Seum, **Amir Hossain Raj**, Shadman Sakib, Tonmoy Hossain, “**A Comparative Study of CNN Transfer Learning Classification Algorithms with Segmentation for COVID-19 Detection from CT Scan Images**”, In 2020 11th International Conference on Electrical and Computer Engineering (ICECE), pp. 234-237. IEEE, 2020.
7. Suvom Shaha, Faisal Muhammad Shah, **Amir Hossain Raj**, Ashek Seum, Saiful Islam, Sifat Ahmed, “**Video Captioning in Bengali With Visual Attention**”, 2022 25th International Conference on Computer and Information Technology (ICCIT 2022).

Workshop

1. B. Panigrahi, **A. H. Raj**, M. Nazeri, and X. Xiao. A Study on Learning Social Robot Navigation with Multimodal Perception. 14th Workshop on Planning, Perception, and Navigation for Intelligent Vehicles, IROS 2024.
2. **Amir Hossain Raj**, Fairuz Nawer Meem , Sadia Afrin Mim. Enhancing Issue Labeling in Open-source Projects, VL/HCC 2025.

Preprint

1. **Amir Hossain Raj**, Dibyendu Das, Xuesu Xiao. Robot Squid Game: Quadrupedal Locomotion for Traversing Narrow Tunnels.
2. Prakhar Mishra, **Amir Hossain Raj**, Xuesu Xiao, Dinesh Manocha. McARL: Morphology-Control-Aware Reinforcement Learning for Generalizable Quadrupedal Locomotion.
3. Prakhar Mishra, **Amir Hossain Raj**, Xuesu Xiao, Dinesh Manocha. HACL: History-Aware Curriculum Learning for Fast Locomotion.

PROFESSIONAL DEVELOPMENT

- Generative Adversarial Networks (GAN) Specialization, Coursera (DeepLearning.AI), Completed 2025
- Deep Learning Specialization, Coursera (DeepLearning.AI), Completed 2020

GRANTS AND ACHIEVEMENTS:

- ACCESS-CI Explore Compute Allocation (Co-PI), NSF ACCESS, 2025.
- “Outstanding Academic Achievement” (based on MS result)
- “Dean’s List of Honor” (based on B.Sc. result)
- Secured **2nd position** in Intra AUST Programming Contest Fall-17